MDPI

Article

Predicting In-Hospital Maternal Mortality and Severe Maternal Morbidity in MIMIC-IV with a Stacked Machine Learning Ensemble under Extreme Class Imbalance

Emmanuel Gutiérrez Jiménez [1,*] 0009-0005-2358-4680 and José Duván Márquez Díaz [2] 0000-0003-0400-5020

[1] System Engineering Department, Universidad del Norte, Barranquilla 081007, Colombia; emmanuelg@uninorte.edu.co
[2] System Engineering Department, Universidad del Norte, Barranquilla 081007, Colombia; jmarquez@uninorte.edu.co
* Correspondence: emmanuelg@uninorte.edu.co; Tel.: +57-310-432-6347

Highlights

What are the main findings?

- Maternal death is too rare in MIMIC-IV to model reliably (41 of 25,736 obstetric admissions, 0.16%): five-fold cross-validated AUC for the stacked ensemble was 0.853 $\pm$ 0.078 but average precision was only 0.164 $\pm$ 0.090.
- Using severe maternal morbidity (SMM; 475 positives, 1.85%) as the outcome gave a markedly more stable task: on a held-out test set the stacked ensemble reached an AUC of 0.913 and an average precision of 0.384 (about 21 times the prevalence), with a recall of 0.79; five-fold cross-validation confirmed it (AUC 0.904 $\pm$ 0.020, AP 0.419 $\pm$ 0.050). Flagging the top 5% of admissions captured 68% of SMM cases.

What are the implications of the main findings?

- In single-center data, SMM is a more suitable modelling target than mortality for maternal risk prediction; point estimates for mortality should not be over-interpreted.
- Removing the diagnosis-group indicators from the mortality model lowered its cross-validated AUC from 0.853 to 0.690, and predicted probabilities were not calibrated: models should be used to rank admissions, not to estimate absolute risk.
- Oversampling (SMOTE) did not improve precision-based metrics, and outlier winsorization avoided discarding 19.5% of deaths and 38.7% of SMM cases that row deletion would have removed.

Abstract

Background: Maternal mortality remains a global health priority, but deaths are so rare in single-center data that standard machine learning (ML) evaluation is unreliable. Severe maternal morbidity (SMM), defined by CDC/HRSA diagnosis and procedure indicators, is about one hundred times more frequent and is widely used as a near-miss proxy. **Methods**: We built two models on the MIMIC-IV electronic health record database from 25,736 obstetric hospital admissions of women aged 12–55 years (observed range 18–55): Model A (death recorded during or after the index admission; 41 events, 0.159%) and Model B (at least one of 20 CDC/HRSA FAD-2023 SMM indicators; 475 events, 1.846%). The pipeline included sentinel-value and implausible-zero cleaning, interquartile-range (IQR$\times$3) winsorization instead of row deletion, skewness-driven imputation, differential scaling, and expert-grouped diagnosis indicators (used in Model A only, to avoid outcome leakage in Model B). Random Forest, XGBoost, Extra Trees, Logistic Regression and a stacked ensemble were tuned by cross-validated grid search on average precision. Stratified five-fold

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Healthcare* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

cross-validation was the primary evidence for Model A and complemented the hold-out evaluation of Model B; SMOTE, a Model A analysis without diagnosis groups, out-of-fold calibration with risk bands, SHAP values and a subgroup audit were added as additional analyses. **Results**: For Model A, the stacked ensemble achieved a cross-validated AUC of 0.853 ± 0.078 and average precision (AP) of 0.164 ± 0.090 (prevalence 0.0016); the single hold-out test contained only eight deaths. For Model B, the stacked ensemble obtained the best hold-out AUC (0.913) and AP (0.384; prevalence 0.018) with recall 0.79 and precision 0.14 at the default threshold; restricted to delivery hospitalizations, AP was 0.646. Cross-validation gave an AUC of 0.904 ± 0.020 and an AP of 0.419 ± 0.050 for Model B, and flagging the top 5% of admissions captured 68% of cases (positive predictive value 0.25). Removing diagnosis groups lowered the cross-validated AUC of Model A from 0.853 to 0.690, predicted probabilities were not calibrated (mean 0.249 against a prevalence of 0.018), and in the subgroup audit the AUC was lowest among admissions with an ICU stay (0.655). SMOTE did not improve AP. **Conclusions**: In this single-center cohort, SMM supports considerably more stable ML evaluation than mortality. Results remain exploratory, because the SMM label is ICD-based, the data are from one institution, and no external validation was performed.

Keywords: maternal mortality; severe maternal morbidity; near miss; MIMIC-IV; stacked ensemble; class imbalance; precision–recall; machine learning; electronic health records

1. Introduction

Maternal mortality remains a major public health problem: the World Health Organization and partner agencies estimated roughly 287,000 maternal deaths worldwide in 2020 [1]; global burden analyses track maternal mortality over time and across regions [2], and a WHO systematic analysis has described its leading causes [3]. In high-resource hospitals, however, maternal death is a very rare event. This rarity is a statistical obstacle for supervised learning: with a prevalence of a few cases per thousand admissions, individual cross-validation folds and test sets contain only a handful of positives, so that a single misclassified patient can change recall by several percentage points, and accuracy and ROC-AUC alone become uninformative [4,5].

A widely used way to mitigate this problem is to model *severe maternal morbidity* (SMM), or "near miss" [6], as a surrogate outcome. The U.S. Centers for Disease Control and Prevention (CDC) define SMM through a set of ICD-10-CM diagnosis and ICD-10-PCS procedure indicators, which the Health Resources and Services Administration (HRSA) distributes in its Federally Available Data (FAD) resource document [7,8]. SMM events occur approximately one hundred times more frequently than in-hospital maternal deaths among delivery and postpartum hospitalizations [9], and administrative diagnosis codes have been shown to carry prognostic information for maternal morbidity [10]. The trade-off is label quality: a validation against medical-record review at a public hospital reported low sensitivity and positive predictive value for ICD-10-based SMM identification [11].

The MIMIC-IV database, from Beth Israel Deaconess Medical Center and successor of MIMIC-III [12], offers de-identified hospital-wide records with laboratory results, charted vital signs, and diagnosis codes [13,14]. Although MIMIC-IV is not an obstetric database, it contains thousands of admissions of women with pregnancy-related diagnoses, which allows a reproducible comparison of the two outcome definitions on the same patients and the same predictors.

Ensemble stacking combines heterogeneous base learners through a meta-learner and often outperforms its components [15,16]. Whether this advantage persists under extreme imbalance, and whether common remedies such as oversampling help [17], is an empirical question. This study therefore makes the following contributions:

- A side-by-side comparison of two outcomes on one obstetric cohort of 25,736 MIMIC-IV admissions: in-hospital-recorded maternal death (Model A) and CDC/HRSA-defined SMM (Model B).
- An imbalance-aware evaluation protocol: probability-based ROC and precision–recall analysis, average precision as the tuning criterion, stratified cross-validation as the primary evidence for the rarest outcome, and an explicit SMOTE sensitivity analysis.
- Explicit leakage control for SMM: diagnosis-group indicators derived from the same ICD codes that define the outcome are excluded from Model B, and a sensitivity analysis restricts evaluation to delivery hospitalizations (the CDC-comparable denominator).
- A winsorization-based outlier policy that retains every positive case.

The remainder of the article is organized as follows. Section 2 reviews related work, Section 3 describes data and methods, Section 4 presents the results, Section 5 discusses findings and limitations, Section 6 considers implications and future work, and Section 7 concludes.

2. Related Work

2.1. Machine learning for maternal morbidity and mortality

A scoping review of 39 studies that applied machine learning to electronic medical record data to predict maternal morbidity and mortality found that boosting methods were the most frequently used approach (18 studies), that cardiovascular and hypertensive outcomes, gestational diabetes and postpartum hemorrhage were the most common targets, that many studies lacked external validation or sensitivity analyses, that only six followed a reporting guideline, and that none described how the models were applied in clinical practice [18]. Maternal death itself is rarely the modelled outcome, and the prediction of perinatal outcomes has been reviewed separately by the authors [19]. Perinatal outcomes, which share data sources and some predictors with maternal ones, have been modelled more often: homogeneous ensembles have been applied to Ethiopian maternal-health and health-insurance data [20], machine-learning models to a Tanzanian birth registry [21], and, in a registry of more than 500,000 pregnancies from six low- and middle-income countries, models for stillbirth and neonatal mortality reached an AUC of 0.71 or less when only prenatal or pre-delivery information was used and above 0.80 in most models that included delivery and postpartum information [22]. This dependence of discrimination on when the predictors are recorded is relevant here, because our predictors also include values measured during the admission.

2.2. Prediction of severe maternal morbidity

Studies of SMM are more numerous because the outcome is more frequent. Gao et al. developed a framework that identified SMM and its risk factors from the electronic health records of 45,858 deliveries at an academic medical center; it reported an AUC of 0.94 compared with 0.80 for a CDC-based model, and surfaced indicators such as electrolyte disturbances, inflammatory-response syndromes and acid–base disorders that are not part of the CDC definition [23]. Li et al. predicted CDC-defined SMM in 261,226 delivery hospitalizations in Maryland (1,976 events; 76 per 10,000 deliveries) with logistic regression and LASSO using linked hospital discharge and hospital-survey data; the best model, a LASSO with 18 variables, reached a cross-validated AUC of 0.796 with low recall in all

models, and the authors reported a roughly twofold unadjusted risk for non-Hispanic Black compared with non-Hispanic White women. They named class imbalance and the lack of external validation as limitations and proposed oversampling techniques such as SMOTE as future work [24]. Diagnosis codes recorded during prenatal care have also been shown to carry prognostic information for maternal morbidity [10].

2.3. Label quality and rare-event methodology

The reliability of SMM labels is a recognized concern: ICD-10-based identification was compared with medical-record review at a public hospital and showed low sensitivity and positive predictive value [11]. At the same time, SMM occurs roughly one hundred times more often than in-hospital maternal death [9], which makes it the usual surrogate when deaths are too few to model. Methodological work on rare events warns that accuracy and ROC-AUC can be uninformative and that precision–recall analysis and explicit reporting of uncertainty are preferable [4,5], and that imbalance corrections may harm calibration without improving discrimination [17].

2.4. Positioning of this study

Relative to the studies above, the present work differs in four respects. First, it uses admission-level vital signs and hospital-wide laboratory values from MIMIC-IV, rather than claims or discharge data, as predictors. Second, it models mortality and SMM on the *same* 25,736 obstetric admissions with the same predictors and pipeline, which allows a direct comparison of the two outcome definitions. Third, it controls outcome leakage explicitly and reports a CDC-comparable delivery-only sensitivity analysis. Fourth, it evaluates the oversampling remedy that Li et al. left as future work [24], using SMOTE strictly inside the cross-validation folds.

3. Materials and Methods

3.1. Data source and study cohort

We used MIMIC-IV hosted on PostgreSQL [13] (version 3.1, released on PhysioNet in October 2024; credentialed access, data downloaded on 18 April 2025). The study cohort comprises hospital admissions of women aged 12–55 years (the observed range was 18–55 years) with at least one obstetric or pregnancy-related ICD diagnosis (diagnosis position ≤ 4), identified by an expanded diagnosis universe combining text matching on the ICD long titles with ICD ranges. An intensive care unit (ICU) stay was *not* required: this is important because 30 of the 41 recorded deaths (73%) occurred in admissions without any ICU stay. The final cohort contains 25,736 admissions, of which 8,857 were identified as delivery hospitalizations. Figure 1 summarizes the workflow.

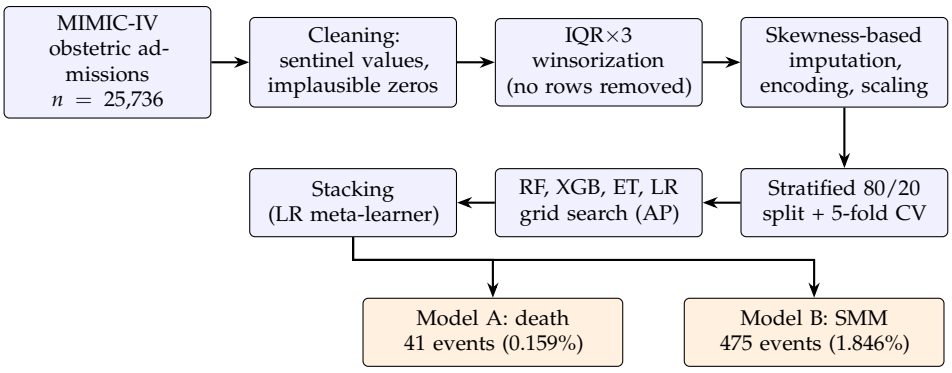


Figure 1. Overview of the pipeline. The same cleaning, preprocessing and learning steps are applied to both outcomes.

3.2. Outcome definitions

Model A (mortality). The outcome is 1 if a hospital death time was recorded during the index admission or in any later admission of the same patient, and 0 otherwise. This yielded 41 events among 25,736 admissions (0.159%).

Model B (SMM). An admission is SMM-positive if it carries at least one ICD-10-CM diagnosis or ICD-10-PCS procedure code matching any of the 20 currently active CDC/HRSA FAD-2023 indicators [7,8,25] (the former indicator on blood-product transfusion, retired by HRSA in 2023, is excluded). The indicators are grouped into seven clinical categories (cardiac, renal, respiratory, hemorrhage, sepsis, other obstetric, other medical). There were 475 SMM-positive admissions (1.846%); among the 8,857 delivery hospitalizations, 252 were positive (2.845%, i.e., 284.5 per 10,000), above the published reference range of 58–88 per 10,000 deliveries [25]. The higher rate is expected, as MIMIC-IV is a tertiary referral center and the expanded denominator also includes non-delivery admissions. The most frequent indicators were acute renal failure (180 admissions), shock (72), ARDS (66), sepsis (62), and disseminated intravascular coagulation (61); 23.4% of positive admissions met indicators from more than one clinical category.

3.3. Predictors and preprocessing

Candidate predictors comprised administrative variables (age at admission, ethnicity group, insurance, language group, admission type, admission location, ICU stay and ICU stay count) and the earliest available value of 27 chart/laboratory variables (vital signs, Glasgow Coma Scale components, hematology and chemistry). For the 13 laboratory variables, values were also taken from hospital-wide laboratory events (not only from ICU chart events), which matters here because most deaths occurred outside the ICU. Preprocessing steps were:

1. **Sentinel values and implausible zeros.** The placeholder 999,999 and physiologically impossible zeros were set to missing.
2. **Outlier handling by winsorization.** For each continuous variable, values beyond $Q_1 - 3\,\mathrm{IQR}$ or $Q_3 + 3\,\mathrm{IQR}$ were clipped to the nearest bound (with clinical lower bounds), so that no admission is deleted. Ordinal scales and variables with $\mathrm{IQR} = 0$ were not clipped. Our first implementation deleted rows with out-of-bound values, as in our previous neonatal pipeline; an audit showed that this removed 8 of 41 deaths (19.5%) in Model A and 184 of 475 SMM cases (38.7%) in Model B, so it was replaced by winsorization.
3. **Imputation.** Median for variables with $|\text{skewness}| > 1$, mode otherwise.
4. **Encoding and scaling.** One-hot encoding of the five categorical groups; `RobustScaler` for variables with $|\text{skewness}| > 1$ and `MinMaxScaler` otherwise. Near-constant ICU-monitoring variables (motor GCS and Richmond Agitation–Sedation Scale; the most common value covered more than 99% of rows) were dropped from Model A.

Diagnosis-group indicators and leakage control. ICD diagnosis codes in the cohort (4,116 distinct codes for Model A) were aggregated into 26 binary diagnostic group indicators (`gdg_m01`–`gdg_m26`) by clinical classification, performed by a panel of four clinical experts (an obstetrician–gynecologist with experience in maternal–fetal medicine, an intensivist with experience in obstetric patients, an internal medicine specialist and an infectious disease specialist) following a structured four-step procedure. First, all ICD codes in the cohort were tabulated by number and percentage of admissions, and this frequency table was the panel's working document. Second, codes frequent enough to identify a well-defined clinical entity were kept as individual groups. Third, less frequent codes were grouped by the experts according to, in order of priority, a shared pathophysiological mechanism, the affected organ system, and similar prognostic relevance; no code was assigned to more

than one group, and disagreements were settled by discussion until consensus. Fourth, the complete set was reviewed in a plenary session for clinical coherence, absence of overlap between groups and sufficient representation in the cohort, and groups that failed these criteria were revised or merged. Codes that could not be assigned to a clinically interpretable group were reserved for a residual clustering step. Two further indicators reserved for residual clustering were constant and removed. Because these indicators are derived from the same coded diagnoses that define SMM, they were *excluded* from Model B; likewise the seven per-category SMM counts, the multiclass label, and the delivery-hospitalization flag were not used as predictors (the latter is kept only to define the evaluation subset). Model A retained the 26 indicators, giving 89 predictors; Model B used 65 predictors. This difference in the predictor sets is considered in Section 5.

Table 1 lists the 26 diagnostic groups. Of the 4,116 distinct ICD codes observed in the cohort (2,192 ICD-10 and 1,924 ICD-9), 4,010 were assigned to one of the 26 groups and 106 were excluded from grouping; twelve groups are obstetric and fourteen cover non-obstetric conditions. A k-means step intended to cluster residual, unassigned codes found no residual codes, so the two reserved cluster indicators were constant and removed.

Table 1. Diagnostic groups used as binary indicators in Model A: description, whether the group is obstetric, and number of distinct ICD codes assigned.

Indicator	Group	Obstetric	ICD codes
gdg_m01	Labor and delivery complications	Yes	225
gdg_m02	Pre-existing condition complicating pregnancy	Yes	167
gdg_m03	Hypertensive disorders of pregnancy	Yes	95
gdg_m04	Other pregnancy-related maternal disorders	Yes	83
gdg_m05	Fetal complications / multiple gestation	Yes	156
gdg_m06	Other puerperal complications	Yes	27
gdg_m07	Placental / amniotic fluid disorders	Yes	43
gdg_m08	Hemorrhage (antepartum/postpartum)	Yes	57
gdg_m09	Infection (obstetric/puerperal)	Yes	117
gdg_m10	Preterm labor / premature rupture of membranes	Yes	35
gdg_m11	Abortion / ectopic or molar pregnancy	Yes	103
gdg_m12	Venous / thromboembolic disorders	Yes	50
gdg_m13	Neoplasms	No	95
gdg_m14	Digestive diseases	No	248
gdg_m15	Endocrine / metabolic / nutritional disorders	No	197
gdg_m16	Cardiovascular diseases	No	231
gdg_m17	Nervous system disorders	No	147
gdg_m18	Blood / immune system diseases	No	107
gdg_m19	Genitourinary diseases (non-obstetric)	No	175
gdg_m20	Respiratory diseases	No	106
gdg_m21	Administrative factors / contact with health services	No	306
gdg_m22	Symptoms, signs and ill-defined findings	No	208
gdg_m23	Infectious and parasitic diseases	No	147
gdg_m24	Injury, poisoning and external causes	No	355
gdg_m25	Other minor comorbidities	No	332
gdg_m26	Mental and behavioral disorders	No	198

3.4. Correlation analysis

Collinearity was reviewed in two passes: a heatmap of all continuous clinical variables and a focused pass over the binary diagnostic indicators, using $|r| \geq 0.70$ as the threshold for a strongly correlated pair. Because the tree-based learners evaluate features independently at each split and the logistic regression is L2-regularized, we retained all correlated variables, since dropping one member of a physiologically paired measurement would discard independently monitored signal without a modelling benefit.

3.5. Learning algorithms and evaluation

We trained Random Forest (RF), XGBoost (XGB), Extra Trees (ET) and L2-regularized Logistic Regression (LR) on an 80/20 stratified split (random seed 42): 20,588 training and 5,148 test admissions, with 33 and 8 deaths in Model A and 380 and 95 SMM cases in Model B. Imbalance was handled through class weighting (`balanced` or `balanced_subsample`; XGB `scale_pos_weight` of 622.9 in A and 53.2 in B). Hyperparameters were tuned with grid search using stratified five-fold cross-validation on the training set with *average precision* as the criterion; grids were deliberately small for Model A. The tuned learners were combined into a stacked ensemble [15] with a class-weighted L2 logistic-regression meta-learner whose inputs were generated by the same five-fold stratified scheme. Implementation used scikit-learn, XGBoost and imbalanced-learn [26–28] with RF [29] and ET [30] base models. The selected hyperparameters of every learner are listed in Table 2.

Table 2. Selected hyperparameters (grid search, five-fold stratified cross-validation on the training split, criterion: average precision). Both forests used `max_features` = sqrt; class weights were `balanced_subsample` for RF and `balanced` for ET and LR.

Model	Model A (mortality)	Model B (SMM)
Random Forest	400 trees; max depth none; min split 5; min leaf 2	300 trees; max depth 20; min split 10; min leaf 4
XGBoost	400 trees; max depth 5; learning rate 0.1; subsample 0.8; column sample 1.0; positive-class weight 622.9	600 trees; max depth 3; learning rate 0.01; subsample 0.8; column sample 0.8; positive-class weight 53.2
Extra Trees	400 trees; max depth none; min split 2; min leaf 1	500 trees; max depth 20; min split 10; min leaf 2
Logistic Regression	L2; $C = 1$; solver lbfgs; 3,000 iterations	L2; $C = 5$; solver lbfgs; 3,000 iterations
Stacking (meta-learner)	L2 logistic regression; $C = 0.1$; solver lbfgs; 5,000 iterations; balanced class weights; five-fold out-of-fold inputs	

ROC and precision–recall (PR) curves [5,31] and the AUC and AP were computed from predicted probabilities of the positive class, while the classification report and confusion matrix used the default 0.5 decision threshold. Since only 8 deaths fall in the Model A test set, the main evidence for Model A is stratified five-fold cross-validation over all 25,736 admissions, reported as mean $\pm$ standard deviation for AUC, AP, recall, F1 and balanced accuracy [32]. Formal sample-size criteria for prediction models depend on the number of candidate predictors, the outcome prevalence and the expected performance [33]; with 41 events and 89 candidate predictors (about 0.5 events per predictor), Model A is unlikely to meet them, which is a further reason to treat its results as exploratory. The cross-validation re-fits the already tuned configurations on the full data set; its estimates are therefore slightly optimistic, as the hyperparameters were selected on the training split. As a sensitivity analysis, SMOTE (sampling strategy 0.2, $k = 5$) was placed inside each training fold of a pipeline to avoid leakage into the validation folds. For Model B, the stacked ensemble was additionally evaluated on the delivery-hospitalization subset of the test set, without re-training. The study follows the TRIPOD+AI reporting guidance where applicable [34] ([**TODO:** attach the TRIPOD+AI checklist as supplementary material]).

3.6. Additional analyses

Five further analyses complement the main evaluation. (i) *Cross-validation for Model B*: the same stratified five-fold protocol applied to Model A was applied to Model B, so that both outcomes are assessed with the same primary evidence; a SMOTE variant was

run in the same way. (ii) *Model A without diagnostic groups*: Model A was re-evaluated by cross-validation after removing the 26 diagnostic-group indicators, keeping the tuned hyperparameters, so that its predictor set matches that of Model B. (iii) *Calibration and risk stratification*: out-of-fold probabilities of the stacked ensemble over the full cohort were used to compute the Brier score [35] and its skill relative to a model that predicts the prevalence, the ratio between mean predicted probability and observed prevalence, a reliability curve with quantile bins [36,37], and "flag the top x%" tables giving recall, positive predictive value and lift. Risk bands were defined from the model's own probability percentiles (low: below the 80th percentile; moderate: 80th to 95th; high: top 5%), because fixed cut-offs are not meaningful at these prevalences, and bootstrap 95% confidence intervals (500 resamples) were computed. (iv) *Explainability*: TreeSHAP values [38,39] of the fitted XGBoost base learner were computed on the Model B test set. (v) *Subgroup audit*: the hold-out performance of the Model B stacked ensemble was described by ethnicity group, insurance, age quartile, ICU stay and delivery versus non-delivery admission, reporting AUC with bootstrap confidence intervals only for subgroups with at least 10 positive and 10 negative admissions. Auditing performance by subgroup responds to documented concerns that algorithms can reproduce or amplify inequities in health care [40,41].

4. Results

4.1. Cohorts and outcomes

Table 3 compares the two prediction tasks. Model B has 11.6 times more events than Model A and 95 rather than 8 positives in the test set.

Table 3. Comparison of the two prediction tasks (same 25,736 admissions).

	Model A (mortality)	**Model B (SMM)**
Positive admissions	41 (0.159%)	475 (1.846%)
Positives in training / test split	33 / 8	380 / 95
Predictors	89 (incl. 26 diagnosis groups)	65 (no diagnosis groups)
Weight for the positive class (XGBoost)	622.9	53.2
Positives retained after outlier handling	41 of 41	475 of 475

4.2. Descriptive analysis of severe maternal morbidity

All 20 CDC/HRSA indicators were observed at least once. Table 4 lists the number of admissions per indicator. Renal indicators were the most frequent category (180 admissions with at least one renal indicator), followed by hemorrhage (131), respiratory (89), other medical (77), sepsis (62), cardiac (61) and other obstetric (56) indicators. When each SMM admission was assigned to the category with the most indicators, the primary categories were renal (132), hemorrhage (115), other medical (71), cardiac (52), other obstetric (38), respiratory (36) and sepsis (31). Most SMM admissions involved a single clinical category (364 of 475, 76.6%), whereas 64 involved two, 27 three, 17 four and 3 five categories (111 admissions, 23.4%, with more than one category). The most frequent pairs were respiratory with hemorrhage (42 admissions), renal with hemorrhage (37), renal with respiratory (28), cardiac with respiratory (26) and hemorrhage with sepsis (20) (Figure 2). The SMM rate was 184.6 per 10,000 admissions in the expanded cohort and 284.5 per 10,000 in delivery hospitalizations, against a published range of 58–88 per 10,000 deliveries.

Table 4. Admissions per CDC/HRSA SMM indicator, grouped by clinical category (an admission may contain several indicators).

Category	Indicator	Admissions
Cardiac	Pulmonary edema / acute heart failure	45
	Conversion of cardiac rhythm	8
	Acute myocardial infarction	7
	Cardiac arrest / ventricular fibrillation	7
	Aneurysm	5
	Heart failure / arrest during surgery or procedure	1
Hemorrhage	Shock	72
	Disseminated intravascular coagulation	61
	Hysterectomy	41
Other medical	Puerperal cerebrovascular disorders	55
	Sickle cell disease with crisis	23
Other obstetric	Air and thrombotic embolism	33
	Eclampsia	16
	Amniotic fluid embolism	4
	Severe anesthesia complications	4
Renal	Acute renal failure	180
Respiratory	Acute respiratory distress syndrome	66
	Ventilation	51
	Temporary tracheostomy	1
Sepsis	Sepsis	62

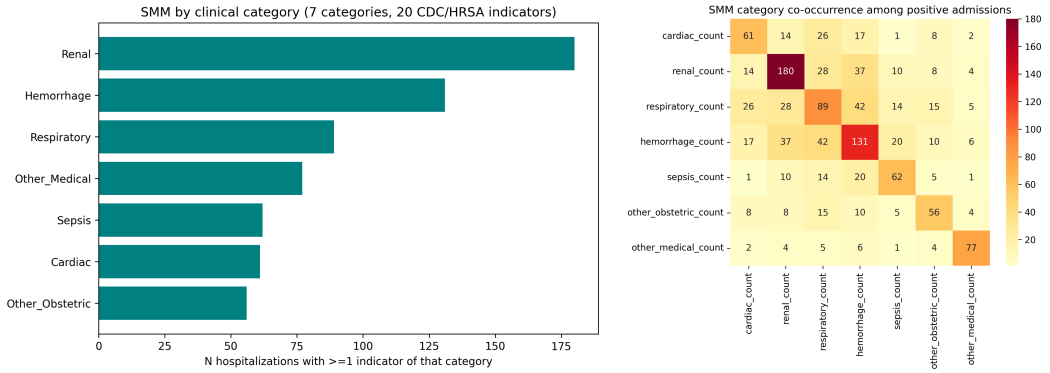


Figure 2. SMM admissions by clinical category (left) and co-occurrence of categories among SMM admissions (right; the diagonal gives the number of admissions per category).

4.3. Correlation structure of the predictors

Ten pairs of clinical variables had $|r| \geq 0.70$ (Table 5), all physiologically expected: ICU stay and ICU stay count, the three blood-pressure measures, the Glasgow Coma Scale components with the Richmond Agitation–Sedation Scale, and hemoglobin with hematocrit. Following the policy in Section 3, no variable was removed for collinearity. Motor GCS and the Richmond scale were dropped from Model A for near-zero variance (most common value in more than 99% of rows), and were retained in Model B.

Table 5. Pairs of clinical variables with $|r| \geq 0.70$ (identical in both models).

Variable 1	Variable 2	r
ICU stay	ICU stay count	0.958
Mean blood pressure	Diastolic blood pressure	0.907
GCS eye opening	GCS verbal response	0.891
GCS eye opening	GCS motor response	0.882
Hemoglobin	Hematocrit	0.839
Mean blood pressure	Systolic blood pressure	0.833
GCS eye opening	Richmond Agitation–Sedation Scale	0.783
GCS verbal response	GCS motor response	0.773
GCS verbal response	Richmond Agitation–Sedation Scale	0.754
GCS motor response	Richmond Agitation–Sedation Scale	0.747

4.4. Model A: mortality

Table 6 and Figure 3 show the cross-validated results. The stacked ensemble had the highest AUC (0.853 ± 0.078) and the highest balanced accuracy and recall (0.752 and 0.611), but its F1 score was only 0.018, and its AP (0.164 ± 0.090) was comparable to that of XGB and ET (0.158 and 0.151) and far below any clinically useful level in absolute terms, although it is roughly one hundred times the prevalence of 0.0016. Fold-to-fold variability was large (the standard deviation of AP reached 0.093 for XGB and 0.090 for the stacked ensemble) and RF produced no positive predictions at the default threshold (recall 0). LR had the lowest AUC and AP (0.737 and 0.023). On the hold-out test set (8 deaths), AUC values ranged from 0.532 (RF) to 0.799 (LR), with the stacked ensemble at 0.778 and an AP of 0.089 (Table 8). Its confusion matrix contained 6 true positives, 2 false negatives, 365 false positives and 4,775 true negatives (recall 0.75, precision 0.016); these point estimates rest on eight patients and must be read as illustrative only.

Table 6. Model A: stratified five-fold cross-validation (mean ± SD) on all 25,736 admissions (41 deaths).

Model	AUC	AP	Recall	F1	Bal. acc.
Random Forest	0.841 ± 0.068	0.110 ± 0.050	0.000 ± 0.000	0.000 ± 0.000	0.500 ± 0.000
XGBoost	0.810 ± 0.081	0.158 ± 0.093	0.142 ± 0.107	$\mathbf{0.183 \pm 0.129}$	0.571 ± 0.054
Extra Trees	0.830 ± 0.070	0.151 ± 0.061	0.050 ± 0.061	0.084 ± 0.104	0.525 ± 0.031
Logistic Regression	0.737 ± 0.077	0.023 ± 0.019	0.439 ± 0.125	0.020 ± 0.007	0.683 ± 0.059
Stacking	$\mathbf{0.853 \pm 0.078}$	$\mathbf{0.164 \pm 0.090}$	$\mathbf{0.611 \pm 0.115}$	0.018 ± 0.003	$\mathbf{0.752 \pm 0.050}$

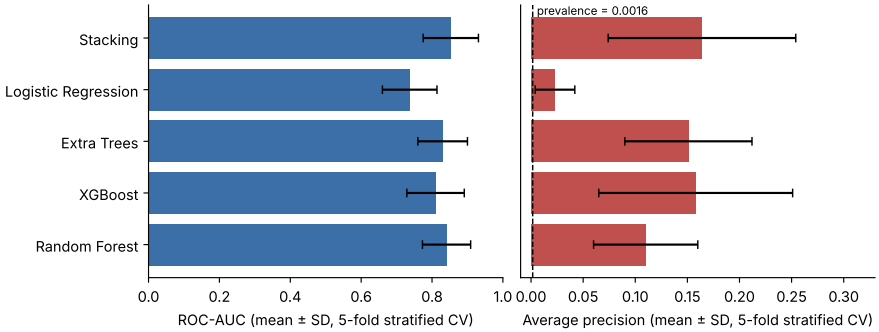


Figure 3. Model A: cross-validated AUC and average precision with standard deviations. The dashed line marks the outcome prevalence (41/25,736).

4.5. Oversampling sensitivity analysis (Model A)

Adding SMOTE inside each training fold increased the AUC of all five models (e.g., stacking 0.853 to 0.882, XGB 0.810 to 0.840, ET 0.830 to 0.865) and the recall of RF and XGB, but not that of the stacked ensemble (0.611 to 0.119) or LR (0.439 to 0.169). AP changed

little for the best models (stacking 0.164 to 0.173, XGB 0.158 to 0.163, ET 0.151 to 0.145), fell for RF (0.110 to 0.075) and rose for LR (0.023 to 0.041) (Table 7, Figure 4). The changes in AUC and AP are all smaller than one standard deviation, whereas the recall of the stacked ensemble dropped markedly. Since oversampling mainly shifts the operating point rather than adding information, and AP did not improve beyond fold-to-fold noise, class weighting was kept as the default.

Table 7. Model A: effect of SMOTE (sampling strategy 0.2) on five-fold cross-validated AUC, AP and recall (mean values).

	AUC		AP		Recall	
Model	Weights	SMOTE	Weights	SMOTE	Weights	SMOTE
Random Forest	0.841	0.862	0.110	0.075	0.000	0.050
XGBoost	0.810	0.840	0.158	0.163	0.142	0.169
Extra Trees	0.830	0.865	0.151	0.145	0.050	0.050
Logistic Regression	0.737	0.776	0.023	0.041	0.439	0.169
Stacking	0.853	0.882	0.164	0.173	0.611	0.119

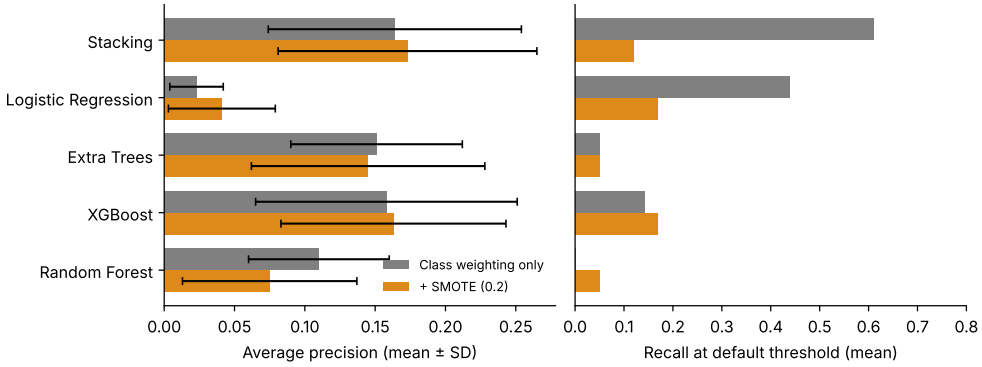


Figure 4. Model A: class weighting versus SMOTE in five-fold cross-validation.

4.6. Model B: severe maternal morbidity

Model B showed a better and much more stable behavior. On the hold-out test set (95 SMM cases), the stacked ensemble was the best model in both AUC (0.913) and AP (0.384), followed by XGB (AUC 0.909, AP 0.358) and RF (AUC 0.900, AP 0.360) (Table 8). The AP of the stacked ensemble is about 21 times the test prevalence (0.018). At the default threshold its confusion matrix showed 75 true positives, 20 false negatives, 473 false positives and 4,580 true negatives (recall 0.79, precision 0.14). Restricted to the 1,802 test admissions that were delivery hospitalizations (58 SMM cases, 3.219%), the same model reached a recall of 0.78, precision of 0.27, F1 of 0.40 for the positive class and an AP of 0.646 (versus 0.384 in the whole test set), with an overall accuracy of 0.93. Figure 5 shows the ROC and PR curves of the stacked ensemble for both models.

Table 8. Hold-out test performance (probabilities of the positive class): AUC and average precision. Model A test set: 5,148 admissions, 8 deaths; Model B test set: 5,148 admissions, 95 SMM cases.

	Model A (mortality)		Model B (SMM)	
Model	AUC	AP	AUC	AP
Random Forest	0.532	0.046	0.900	0.360
XGBoost	0.736	0.139	0.909	0.358
Extra Trees	0.658	0.073	0.900	0.315
Logistic Regression	0.799	0.150	0.889	0.339
Stacking	0.778	0.089	**0.913**	**0.384**

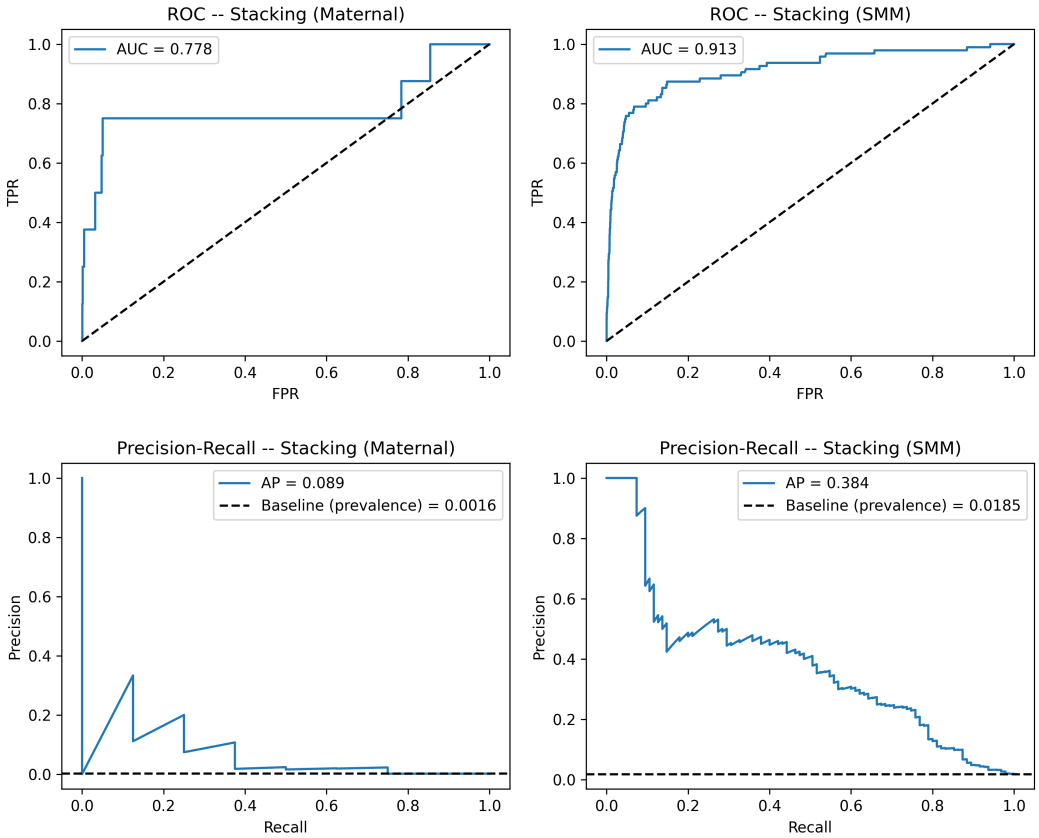


Figure 5. ROC (top) and precision–recall (bottom) curves of the stacked ensemble on the hold-out set. Left: Model A; right: Model B.

4.7. Predictor importance

Figure 6 shows the 30 most important predictors of the tuned Random Forest for each model. For Model A, age at admission ranked first, followed by hematocrit, hemoglobin, the diagnosis-group indicator `gdg_m13`, and creatinine. For Model B, the top positions were occupied by laboratory values: creatinine, glucose, bicarbonate, magnesium, and hemoglobin. Impurity-based importances are descriptive of the fitted forests and not evidence of causal effects; moreover, the importance threshold used for feature screening was computed but not applied, so that all predictors were retained in every model.

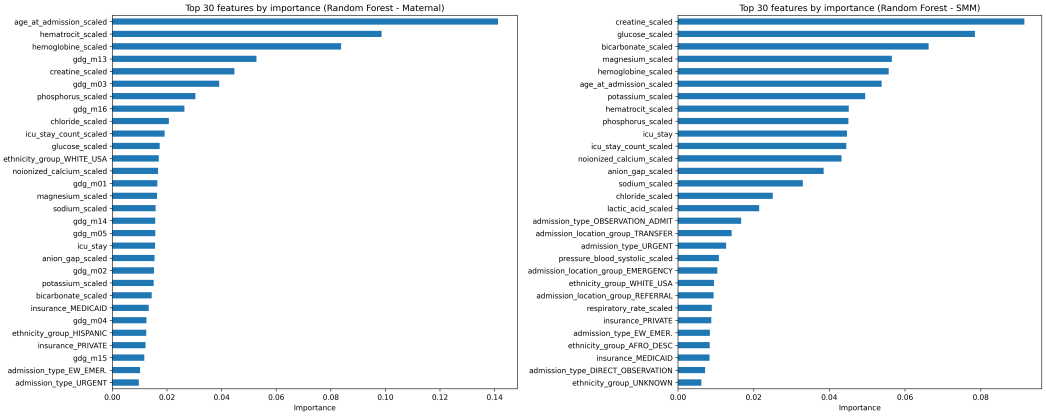


Figure 6. Top 30 Random Forest feature importances. Left: Model A; right: Model B.

4.8. Additional analyses

Cross-validation for Model B.

Over the full cohort (475 SMM cases), the same five-fold protocol gave results that were consistent with the hold-out evaluation and considerably less dispersed than those of Model A (Table 9, Figure 7). The stacked ensemble had the highest AUC and AP (0.904 $\pm$ 0.020 and 0.419 $\pm$ 0.050, about 23 times the prevalence), with XGBoost nearly identical (0.901 and 0.413); the standard deviations of the AUC across folds were 0.017–0.029, against 0.068–0.081 for Model A. The stacked ensemble was also the best model on the hold-out set (AUC 0.913, AP 0.384), although the differences among the top models are smaller than one standard deviation of AP. The operating point at the default threshold differed markedly between models: the stacked ensemble and XGBoost reached recall of 0.800 and 0.758 at the cost of low F1 (0.234 and 0.264), whereas Random Forest was more conservative (recall 0.467, F1 0.444).

Table 9. Model B: stratified five-fold cross-validation (mean $\pm$ SD) on all 25,736 admissions (475 SMM cases).

Model	AUC	AP	Recall	F1
Random Forest	0.890 $\pm$ 0.028	0.368 $\pm$ 0.037	0.467 $\pm$ 0.034	0.444 $\pm$ 0.032
XGBoost	0.901 $\pm$ 0.017	0.413 $\pm$ 0.038	0.758 $\pm$ 0.040	0.264 $\pm$ 0.011
Extra Trees	0.885 $\pm$ 0.023	0.359 $\pm$ 0.042	0.560 $\pm$ 0.041	0.362 $\pm$ 0.035
Logistic Regression	0.876 $\pm$ 0.029	0.342 $\pm$ 0.048	0.749 $\pm$ 0.074	0.233 $\pm$ 0.018
Stacking	**0.904** $\pm$ 0.020	**0.419** $\pm$ 0.050	0.800 $\pm$ 0.040	0.234 $\pm$ 0.013

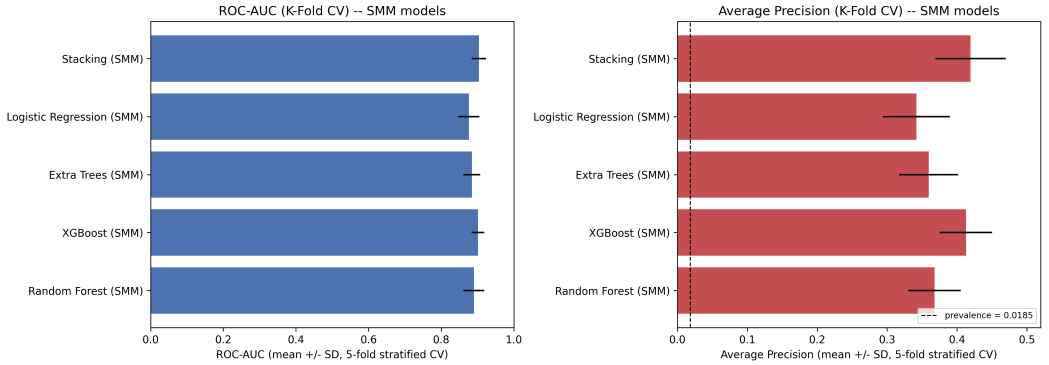


Figure 7. Model B: five-fold cross-validated AUC and AP (mean and standard deviation); the dashed line is the prevalence.

Oversampling sensitivity analysis (Model B).

SMOTE did not improve the precision-based metrics of Model B either (Table 10, Figure 8). The AP of the stacked ensemble went from 0.419 to 0.406 and that of XGBoost from 0.413 to 0.350, while the AUC of the stacked ensemble went from 0.904 to 0.896. Only Random Forest (AUC 0.890 to 0.899) and LR (AP 0.342 to 0.357) showed small gains. SMOTE shifted the operating point instead: the recall of XGBoost rose from 0.758 to 0.964, but with an F1 of 0.047, that is, almost all admissions were flagged, whereas the recall of the stacked ensemble fell from 0.800 to 0.636. As in Model A, class weighting was kept as the default.

Table 10. Model B: effect of SMOTE (sampling strategy 0.2) on five-fold cross-validated AUC, AP and recall (mean values).

	AUC		AP		Recall	
Model	Weights	SMOTE	Weights	SMOTE	Weights	SMOTE
Random Forest	0.890	0.899	0.368	0.366	0.467	0.539
XGBoost	0.901	0.892	0.413	0.350	0.758	0.964
Extra Trees	0.885	0.875	0.359	0.349	0.560	0.535
Logistic Regression	0.876	0.825	0.342	0.357	0.749	0.554
Stacking	0.904	0.896	0.419	0.406	0.800	0.636

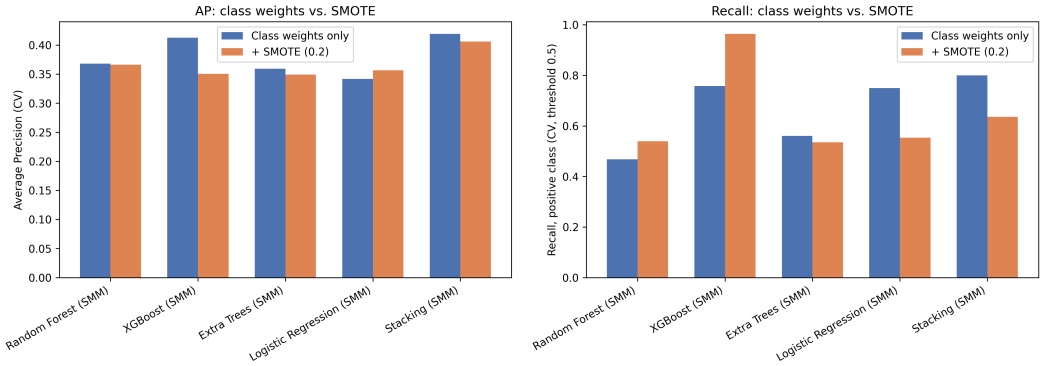


Figure 8. Model B: class weighting versus SMOTE in five-fold cross-validation (left: AP; right: recall at threshold 0.5).

Model A without diagnostic groups.

Table 11 compares the cross-validated performance of Model A with and without the 26 diagnostic-group indicators. Removing them lowered the AUC of every model (stacking 0.853 to 0.690, RF 0.841 to 0.728, XGB 0.810 to 0.721, ET 0.830 to 0.781, LR 0.737 to 0.688) and brought AP down to 0.010–0.053 for all models (stacking 0.164 to 0.012, against a prevalence of 0.0016), with wider fold-to-fold dispersion (the standard deviation of the stacking AUC rose from 0.078 to 0.144). On the hold-out set, the stacked ensemble fell from an AUC of 0.778 and an AP of 0.089 to 0.764 and 0.008 (eight positives; illustrative only). Most of the discriminative signal of Model A therefore comes from the diagnosis groups, which are derived from the ICD codes of the admission itself.

Table 11. Model A: cross-validated AUC and AP with and without the diagnostic-group indicators (hyperparameters kept; mean values).

	AUC		AP	
Model	With	Without	With	Without
Random Forest	0.841	0.728	0.110	0.010
XGBoost	0.810	0.721	0.158	0.053
Extra Trees	0.830	0.781	0.151	0.024
Logistic Regression	0.737	0.688	0.023	0.016
Stacking	0.853	0.690	0.164	0.012

Calibration and risk stratification of Model A.

Using the out-of-fold probabilities of the stacked ensemble over the full cohort (41 deaths), the AUC was 0.835 (bootstrap 95% confidence interval 0.754–0.904) and the AP was 0.112 (0.037–0.243). As expected after class weighting, the predicted probabilities were far above the observed frequency: the mean predicted probability was 0.369 against an observed prevalence of 0.0016 (ratio 231.7), the Brier score was 0.152 (0.0016 for a model that always predicts the prevalence; Brier skill score −94.6), and the reliability curve lay

below the diagonal over its whole range (Figure 9). Rank-based measures do not depend on calibration (Table 12). Flagging the top 5% of admissions (1,287) captured 19 of the 41 deaths (recall 0.463, 95% confidence interval 0.303–0.615; positive predictive value 0.015, 0.009–0.022; lift 9.3). With percentile-based risk bands, the observed mortality was 0.05% in the low band (below the 80th percentile; 10 deaths among 20,588 admissions), 0.31% in the moderate band (80th to 95th percentile; 12 of 3,861) and 1.48% in the high band (top 5%; 19 of 1,287), a roughly 30-fold gradient between the high and the low band. On the hold-out set, 5 of the 8 deaths fell in the high band, 1 in the moderate band and 2 in the low band.

Table 12. Model A: "flag the top x%" table from out-of-fold predictions of the stacked ensemble (41 deaths among 25,736 admissions).

Top x% flagged	Admissions	Deaths captured	Recall	PPV	Lift
0.5%	129	10	0.244	0.0775	48.7
1%	257	13	0.317	0.0506	31.8
2%	515	15	0.366	0.0291	18.3
5%	1,287	19	0.463	0.0148	9.3
10%	2,574	24	0.585	0.0093	5.9
20%	5,147	31	0.756	0.0060	3.8

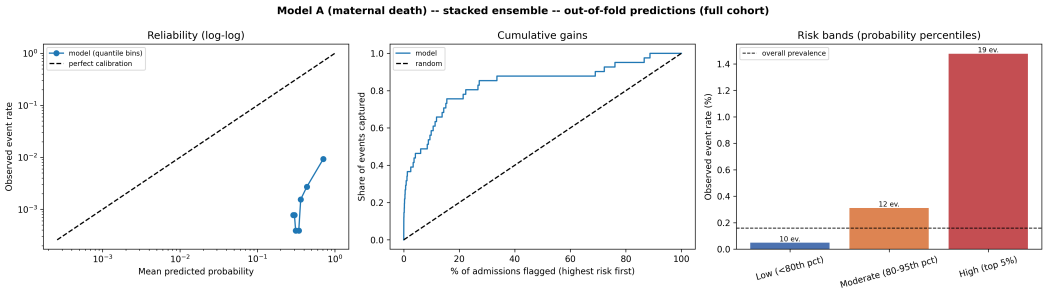


Figure 9. Model A, stacked ensemble, out-of-fold predictions over the full cohort: reliability curve (log–log scale), cumulative gains and observed mortality by risk band.

Calibration and risk stratification of Model B.

The out-of-fold probabilities of the stacked ensemble over the full cohort (475 SMM cases) gave an AUC of 0.902 (bootstrap 95% confidence interval 0.884–0.920) and an AP of 0.396 (0.349–0.454). The probabilities were again much higher than the observed frequency: the mean predicted probability was 0.249 against a prevalence of 0.0185 (ratio 13.5), the Brier score was 0.092 (0.018 for a model that always predicts the prevalence; Brier skill score −4.10), and the reliability curve lay below the diagonal (Figure 10). The hold-out set showed the same pattern (mean predicted probability 0.252, ratio 13.6; Brier score 0.093). The probabilities should therefore be used to rank admissions and not as absolute risks. In ranking terms (Table 13), flagging the top 5% of admissions (1,287) captured 324 of the 475 SMM cases (recall 0.682, bootstrap 95% confidence interval 0.643–0.719; positive predictive value 0.252, 0.226–0.277; lift 13.6), and flagging the top 1% (257) already captured 151 cases with a positive predictive value of 0.588. With percentile-based risk bands, the observed SMM rate was 0.34% in the low band (70 cases among 20,588 admissions), 2.10% in the moderate band (81 of 3,861) and 25.2% in the high band (324 of 1,287), a gradient of about 74 times between the high and the low band; the high band contained 68% of all cases and the low band 15%. In the hold-out set, 64 of the 95 cases fell in the high band (observed rate 24.8%), 19 in the moderate band (2.5%) and 12 in the low band (0.29%); recall and positive predictive value in the top 5% were 0.674 (0.589–0.780) and 0.249 (0.195–0.309). Percentile-based bands were used because the probability cut-offs (0.27 and 0.80 for the

moderate and high bands) are specific to this model and are not meaningful as absolute risks.

Table 13. Model B: "flag the top x%" table from out-of-fold predictions of the stacked ensemble (475 SMM cases among 25,736 admissions).

Top x% flagged	Admissions	Cases captured	Recall	PPV	Lift
0.5%	129	79	0.166	0.612	33.2
1%	257	151	0.318	0.588	31.8
2%	515	238	0.501	0.462	25.0
5%	1,287	324	0.682	0.252	13.6
10%	2,574	372	0.783	0.145	7.8
20%	5,147	405	0.853	0.079	4.3

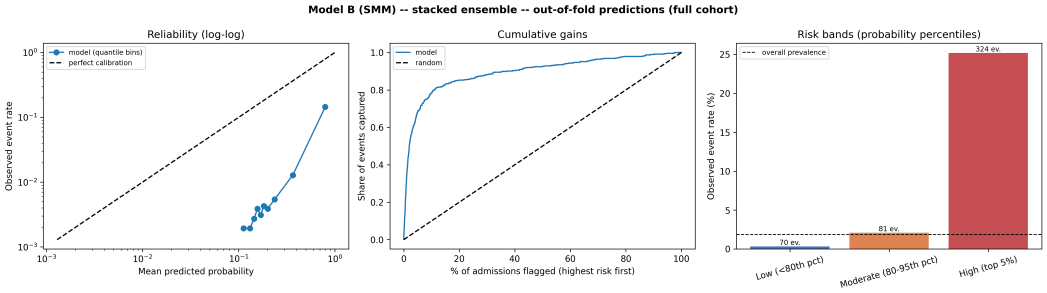


Figure 10. Model B, stacked ensemble, out-of-fold predictions over the full cohort: reliability curve (log–log scale), cumulative gains and observed SMM rate by risk band.

Explainability and subgroup audit.

The mean absolute SHAP values of the XGBoost base learner on the test set were highest for creatinine (0.28), glucose (0.24), ICU stay (0.24), bicarbonate (0.18), lactic acid (0.16), magnesium (0.14), age (0.13) and hemoglobin (0.13), followed by hematocrit, observation and urgent admission types, and transfer admissions (0.08–0.09) (Figure 11). The ranking agrees with the Random Forest importances (renal and metabolic laboratory values at the top) and adds ICU stay and lactic acid. These are descriptions of the fitted model and not causal effects; moreover, ICU stay and lactate may follow the morbid event rather than precede it, so their weight should be read with the timing caveat discussed below.

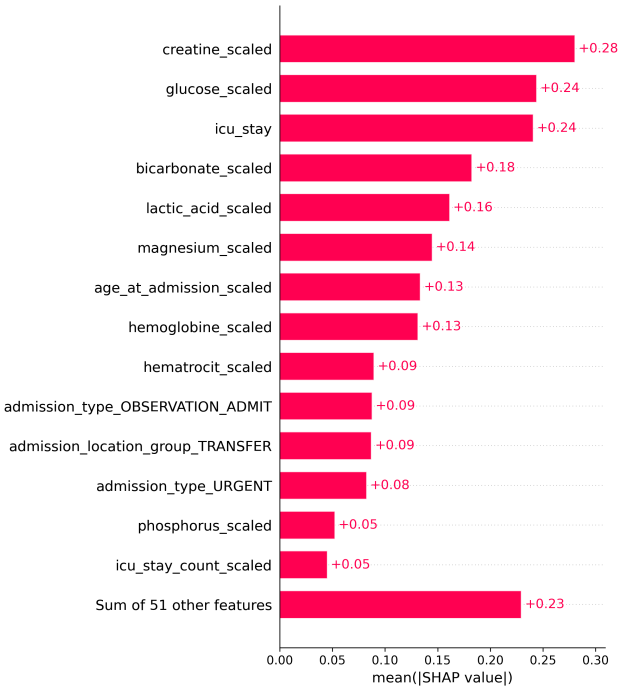


Figure 11. Model B: mean absolute SHAP value of the 15 most influential predictors of the XGBoost base learner (test set). Variable names are those of the notebook; the suffix `_scaled` denotes the scaled version of each variable.

On the hold-out set, the AUC of the stacked ensemble was between 0.86 and 0.95 in nearly all subgroups with at least 10 positive admissions (Table 14); the confidence intervals were wide and overlapped for ethnicity groups, insurance categories and age quartiles, so no difference between them can be asserted. Subgroups with fewer than 10 positives (Asian, Hispanic, Indian-American, European-White and Pacific-Islander groups; Medicare, other insurance and unclassified insurance) were not evaluated. The clear exception was the 94 admissions with an ICU stay, among which 36 (38%) were SMM cases and the AUC was 0.655 (0.539–0.767): the model flagged 99% of them, so ICU stay was a strong predictor of SMM by itself but the model did not separate cases from non-cases inside this group. Performance was similar in delivery (AUC 0.920) and non-delivery hospitalizations (0.913), although the AP was three times higher in the former (0.646 against 0.221), reflecting its higher prevalence. All subgroup estimates are descriptive, are based on a single test split, and do not constitute a formal fairness assessment.

Table 14. Model B: hold-out performance of the stacked ensemble by subgroup. Only subgroups with at least 10 positive admissions are shown. Age groups correspond to the quartiles of the scaled age, approximately ≤28, 29–32, 33–36 and ≥37 years.

Dimension	Subgroup	Admissions	SMM cases	AUC (95% CI)	AP
Ethnicity group	AFRO_DESC	1,285	28	0.866 (0.769–0.946)	0.304
	UNKNOWN	619	17	0.944 (0.865–0.984)	0.532
	WHITE_USA	2,092	39	0.921 (0.868–0.965)	0.355
Insurance	Medicaid	1,731	32	0.902 (0.820–0.967)	0.430
	Private	3,194	60	0.923 (0.878–0.963)	0.405
Age	≤28 years	1,312	20	0.884 (0.773–0.968)	0.351
	29–32 years	1,271	18	0.940 (0.862–0.986)	0.317
	33–36 years	1,333	29	0.953 (0.917–0.982)	0.473
	≥37 years	1,232	28	0.860 (0.763–0.949)	0.404
ICU stay	Yes	94	36	0.655 (0.539–0.767)	0.609
	No	5,054	59	0.873 (0.815–0.929)	0.219
Hospitalization	Delivery	1,802	58	0.920 (0.874–0.963)	0.646
	Non-delivery	3,346	37	0.913 (0.851–0.960)	0.221

5. Discussion

Mortality is not a modelling target at this sample size. With 41 deaths, cross-validated AUC values of 0.74–0.85 coexisted with AP of 0.02–0.16, with standard deviations of AP of the same order as the means, and with hold-out conclusions supported by eight deaths. A ranking of the five algorithms is therefore not reliable: for example, LR achieved the best hold-out AUC and AP (0.799 and 0.150) but the lowest cross-validated AUC and AP (0.737 and 0.023), whereas RF had the lowest hold-out AUC (0.532) but the second-highest cross-validated AUC (0.841). Our results support the recommendation to report precision–recall metrics and fold-level dispersion for rare outcomes [4,5].

SMM is a more stable surrogate, but a noisy one. A positive class eleven times larger produced hold-out AUC values of 0.889–0.913 with a narrow spread across algorithms, a stacked ensemble that was best on both AUC and AP on the hold-out set and in cross-validation (0.904 ± 0.020 and 0.419 ± 0.050, with fold-to-fold standard deviations of the AUC about one quarter of those of Model A), and a precision–recall trade-off that can be discussed clinically. In ranking terms, the top 5% of admissions held 68% of SMM cases with a positive predictive value of 0.25 (lift 13.6). Two limitations should be stated plainly. First, the SMM label is ICD-based; ICD-10 coding showed low sensitivity and positive predictive value in a medical-record validation study [11], so our positives are an administrative proxy, and the model may partly learn coding and care-intensity patterns. This is also likely to explain the high SMM rate in delivery hospitalizations. Second, the leading predictors of Model B were creatinine, glucose, bicarbonate and magnesium, which relate to renal and metabolic derangement; since acute renal failure is the most frequent indicator (180 admissions), part of the signal reflects laboratory abnormalities that co-occur with the outcome definition rather than anticipate it. Using the earliest available value mitigates but does not remove this concern, and prospective timing of predictors relative to the event was not enforced. The SHAP analysis and the subgroup audit point in the same direction: ICU stay and lactic acid ranked among the most influential predictors, and the AUC was lowest (0.655) in the 94 test admissions with an ICU stay, where 36 were SMM cases and the model flagged 99%; ICU stay is therefore itself a strong marker of SMM, but the model does not separate cases from non-cases among the critically ill. Differences across ethnicity, insurance and age groups were not distinguishable from sampling noise (overlapping intervals based on 17–60 positives per subgroup), and no claim about equity can be made from them.

Oversampling and outlier policy. SMOTE did not improve the precision-based metrics in either model (stacking AP 0.164 to 0.173 in Model A and 0.419 to 0.406 in Model B), and it shifted the operating point in unpredictable ways (XGBoost recall in Model B 0.758 to 0.964 with an F1 of 0.047), in line with evidence that imbalance corrections can harm calibration and do not necessarily improve discrimination [17]. Replacing row deletion by winsorization preserved all positive cases; retaining them matters more than the geometric tidiness of the feature distribution, since deleting extreme laboratory values removes precisely the sickest patients.

Limitations. (i) Single-center data and no external validation. (ii) Model A includes the 26 diagnosis-group indicators whereas Model B excludes them; the two models are therefore not strictly comparable in predictor sets. Removing the indicators from Model A lowered its cross-validated AUC from 0.853 to 0.690 and its AP from 0.164 to 0.012, so its performance with the groups depends on variables derived from the admission's own diagnoses, which may reflect the clinical course leading to death rather than anticipate it. (iii) Imputation values and scalers were fitted before the train/test split, and the cross-validation re-fits hyperparameters tuned on the training split, which can make estimates slightly optimistic. (iv) The ICD-based SMM label and the laboratory and ICU-related predictors may be partly concurrent with the event (the timing of predictors relative to the event was not enforced). (v) The mortality outcome captures only deaths recorded by the hospital; post-discharge deaths are not observed. (vi) Metrics at the default 0.5 threshold are shown for transparency; calibration was assessed but not corrected: probabilities were 13.5 times (Model B) and 232 times (Model A) higher than the observed prevalence, no threshold optimization was carried out, and predicted probabilities should not be interpreted as absolute risks, since class weighting and tree ensembles are known to yield poorly calibrated probabilities [17,36,42]. (vii) MIMIC-IV is not an obstetric registry and may differ from the population of obstetric services in other health systems. The models are research prototypes and should not guide individual clinical decisions.

6. Implications and Future Work

Implications for practice. The results do not support using either model for individual decisions. They do indicate how a maternal risk tool should be designed: as a ranking of admissions for review rather than a classifier at a default threshold, with the share of admissions flagged chosen by a clinical panel in view of the recall and precision trade-off, and with absolute risks recalibrated before display. Any operating point or action triggered by such a tool in obstetric care requires validation by an authorized human decision-maker.

Surveillance rather than prediction. Because the SMM categories are clinically distinct (renal, hemorrhagic, respiratory and others), and almost a quarter of SMM admissions involve more than one category, a hospital could use the same features to monitor SMM burden by category and to review cases, even if individual prediction remains imperfect.

Ethical considerations and governance. Four issues deserve explicit mention. First, the data come from a single academic medical center in the United States and were accessed under the PhysioNet credentialed data use agreement, which prohibits re-identification and the sharing of raw data with non-credentialed parties; any reuse of these models with other patient data requires its own authorization and ethical review. Second, regarding algorithmic bias, the subgroup audit (Table 14) was descriptive and could not evaluate several ethnicity and insurance groups for lack of events; the lowest AUCs were observed in the AFRO_DESC group (0.866) and, above all, among admissions with an ICU stay (0.655), and predictions may be least reliable for the groups that are already under-represented in the training data [40,41]. Third, regarding transparency, SHAP values describe associations learned by the model and not causal mechanisms, and a leading predictor may be a

proxy for an unmeasured factor; a model card documenting the intended use, the data, the performance by subgroup and the known limitations [43] would be a minimum requirement for any release of these models. Fourth, regarding human oversight, the models are research prototypes and should not be used in clinical care; any future tool would have to act as a non-binding aid whose alerts clinicians can override and document, be preceded by prospective validation and regulatory review in the applicable jurisdiction, and be monitored after deployment for drift in discrimination and calibration (for example with a rolling AUC and Brier score) overall and by subgroup.

Future work. Five directions follow from the limitations. (i) External validation on other hospital cohorts, in particular obstetric services, to test whether the laboratory-driven signal generalizes; where patient-level data cannot be pooled, federated learning could allow multicenter training and validation without sharing records [44,45]. (ii) Validation of the ICD-based SMM label against medical-record review in a sample, to estimate its real sensitivity and positive predictive value in this cohort. (iii) Restricting predictors to values recorded before the event, which would turn the SMM model into a genuine early-warning model. (iv) Recalibration of the stacked ensemble (the present analysis only documents its miscalibration) and decision-curve analysis [46]. (v) A secondary multiclass model that predicts the primary SMM category among positive admissions, which answers the clinically distinct question of which complication is most likely once SMM occurs.

7. Conclusions

In 25,736 obstetric admissions of MIMIC-IV, in-hospital-recorded maternal death (41 events) could not be modelled with reliable precision, whereas SMM (475 events) allowed a stable evaluation, with a stacked ensemble reaching a hold-out AUC of 0.913 and AP of 0.384, confirmed by cross-validation (0.904 and 0.419); the top 5% of admissions by predicted probability contained 68% of SMM cases. The apparent discrimination for mortality depended largely on diagnosis-group indicators, the probabilities of both models were not calibrated, and the performance was lowest among admissions with an ICU stay. Class weighting was preferable to SMOTE, and winsorization protected positive cases from deletion. Future work should validate these models on independent cohorts, recalibrate and threshold them for a decision-support use case, and evaluate the secondary multiclass task of predicting the SMM category.

Author Contributions: Conceptualization, E.G.J. and J.D.M.D.; methodology, E.G.J.; software, E.G.J.; validation, E.G.J. and J.D.M.D.; formal analysis, E.G.J.; data curation, E.G.J.; writing—original draft preparation, E.G.J.; writing—review and editing, J.D.M.D.; supervision, J.D.M.D. All authors have read and agreed to the published version of the manuscript. [**TODO:** confirm contributions]

Funding: This research was supported by the Becas de Excelencia Doctoral del Bicentenario program, funded by the Fondo de Ciencia, Tecnología e Innovación del Sistema General de Regalías (FCTeI-SGR) and administered by the Ministerio de Ciencia, Tecnología e Innovación (Minciencias) of Colombia. This support was provided through BPIN 2020000100287, through Universidad del Norte, Barranquilla, Colombia, as executing institution.

Institutional Review Board Statement: Ethical review and approval were waived for this study owing to the retrospective, fully de-identified, secondary-use nature of the MIMIC-IV database. The original collection and de-identification of the data were reviewed and approved by the Institutional Review Board of the Beth Israel Deaconess Medical Center (BIDMC), which granted a waiver of informed consent for the creation and sharing of the database [13].

Informed Consent Statement: Patient consent was waived by the Institutional Review Board of the Beth Israel Deaconess Medical Center for the original collection of the MIMIC-IV database, owing to

its retrospective and fully de-identified nature [13]. No participants were directly recruited by the authors.

Data Availability Statement: The clinical data used in this study are available from the MIMIC-IV database but restrictions apply to their availability: they were used under a data use agreement and are therefore not publicly redistributable. Data are available via PhysioNet [14] to any researcher who completes the required training and signs the PhysioNet Credentialed Health Data Use Agreement (https://physionet.org/content/mimiciv/). The SQL scripts and notebooks for cohort extraction, preprocessing and modelling are available at [**TODO:** insert public repository URL].

Acknowledgments: The authors thank the MIMIC-IV team at the Laboratory for Computational Physiology (MIT) and Beth Israel Deaconess Medical Center for maintaining and providing open access to the database. [**TODO:** add any acknowledgments, e.g., clinical reviewers of the diagnosis grouping]

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Abbreviations

AP	Average Precision
AUC	Area Under the Receiver Operating Characteristic Curve
BIDMC	Beth Israel Deaconess Medical Center
CDC	Centers for Disease Control and Prevention
CV	Cross-Validation
ET	Extra Trees
FAD	Federally Available Data
HRSA	Health Resources and Services Administration
ICD	International Classification of Diseases
ICU	Intensive Care Unit
IQR	Interquartile Range
LR	Logistic Regression
MIMIC	Medical Information Mart for Intensive Care
ML	Machine Learning
PR	Precision–Recall
RF	Random Forest
ROC	Receiver Operating Characteristic
SMM	Severe Maternal Morbidity
SMOTE	Synthetic Minority Over-sampling Technique
XGB	XGBoost (Extreme Gradient Boosting)

References

1. World Health Organization. Trends in maternal mortality 2000 to 2020: estimates by WHO, UNICEF, UNFPA, World Bank Group and UNDESA/Population Division. Technical report, World Health Organization, Geneva, 2023.
2. Kassebaum, N.J.; et al. Global, regional, and national levels of maternal mortality, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *The Lancet* **2016**, *388*, 1775–1812.
3. Say, L.; Chou, D.; Gemmill, A.; Tunçalp, Ö.; Moller, A.B.; Daniels, J.; Gülmezoglu, A.M.; Temmerman, M.; Alkema, L. Global causes of maternal death: a WHO systematic analysis. *The Lancet Global Health* **2014**, *2*, e323–e333. https://doi.org/10.1016/S2214-109X(14)70227-X.
4. Marshall, B.D.L. Outcome class imbalance and rare events: An underappreciated complication for overdose risk prediction modeling. *Addiction* **2023**, *118*, 1167–1174. https://doi.org/10.1111/add.16133.
5. Saito, T.; Rehmsmeier, M. The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLoS ONE* **2015**, *10*, e0118432. https://doi.org/10.1371/journal.pone.0118432.
6. Say, L.; Souza, J.P.; Pattinson, R.C. Maternal near miss—towards a standard tool for monitoring quality of maternal health care. *Best Practice & Research Clinical Obstetrics & Gynaecology* **2009**, *23*, 287–296. https://doi.org/10.1016/j.bpobgyn.2009.01.007.

7. Centers for Disease Control and Prevention. Severe Maternal Morbidity in the United States. https://www.cdc.gov/maternal-infant-health/php/severe-maternal-morbidity/, 2023.
8. Health Resources and Services Administration, Maternal and Child Health Bureau. Federally Available Data (FAD) Resource Document. https://mchb.tvisdata.hrsa.gov/Home/Resources, 2023.
9. Callaghan, W.M.; Creanga, A.A.; Kuklina, E.V. Severe Maternal Morbidity Among Delivery and Postpartum Hospitalizations in the United States. *Obstetrics & Gynecology* **2012**, *120*, 1029–1036. https://doi.org/10.1097/AOG.0b013e31826d60c5.
10. Clapp, M.A.; Perlis, R.H.; et al. Prenatal diagnosis codes predict maternal morbidity, aid in risk stratification. *Journal of Perinatology* **2021**.
11. Boulet, S.L.; Stanhope, K.K.; Valdez-Sinon, A.N.; Vuncannon, D.; Preslar, J.; Bergbower, H.; Gray, B.; Gathoo, A.; Hansen, N.; Andre, K.; et al. Validation of ICD-10 Codes for Severe Maternal Morbidity at Delivery in a Public Hospital. *Epidemiology* **2024**, *35*, 506–511. https://doi.org/10.1097/EDE.0000000000001743.
12. Johnson, A.E.W.; Pollard, T.J.; Shen, L.; Lehman, L.w.H.; Feng, M.; Ghassemi, M.; Moody, B.; Szolovits, P.; Celi, L.A.; Mark, R.G. MIMIC-III, a freely accessible critical care database. *Scientific Data* **2016**, *3*, 160035. https://doi.org/10.1038/sdata.2016.35.
13. Johnson, A.E.W.; Bulgarelli, L.; Shen, L.; Gayles, A.; Shammout, A.; Horng, S.; Pollard, T.J.; Hao, S.; Moody, B.; Gow, B.; et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data* **2023**, *10*, 1.
14. Goldberger, A.L.; Amaral, L.A.N.; Glass, L.; Hausdorff, J.M.; Ivanov, P.C.; Mark, R.G.; Mietus, J.E.; Moody, G.B.; Peng, C.K.; Stanley, H.E. PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals. *Circulation* **2000**, *101*, e215–e220.
15. Wolpert, D.H. Stacked generalization. *Neural Networks* **1992**, *5*, 241–259.
16. Hastie, T.; Tibshirani, R.; Friedman, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2 ed.; Springer: New York, NY, 2009.
17. van den Goorbergh, R.; van Smeden, M.; Timmerman, D.; Van Calster, B. The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression. *Journal of the American Medical Informatics Association* **2022**, *29*, 1525–1534. https://doi.org/10.1093/jamia/ocac093.
18. Vasudevan, L.; Kibria, M.G.; Kucirka, L.M.; Shieh, K.; Wei, M.; Masoumi, S.; Balasubramanian, S.; Victor, A.; Conklin, J.L.; Gurcan, M.N.; et al. Machine Learning Models to Predict Risk of Maternal Morbidity and Mortality From Electronic Medical Record Data: Scoping Review. *Journal of Medical Internet Research* **2025**, *27*, e68225. https://doi.org/10.2196/68225.
19. Gutiérrez Jiménez, E.; Márquez Díaz, J.D. Artificial Intelligence for Neonatal and Perinatal Mortality Prevention: A Systematic Review of Machine Learning and Deep Learning Applications. *Healthcare* **2026**, *14*, 2339. https://doi.org/10.3390/healthcare14152339.
20. Bogale, D.S.; Abuhay, T.M.; Dejene, B.E. Predicting perinatal mortality based on maternal health status and health insurance service using homogeneous ensemble machine learning methods. *BMC Medical Informatics and Decision Making* **2022**, *22*, 341. https://doi.org/10.1186/s12911-022-02084-1.
21. Mboya, I.B.; Mahande, M.J.; Mohammed, M.; Obure, J.; Mwambi, H.G. Prediction of perinatal death using machine learning models: a birth registry-based cohort study in northern Tanzania. *BMJ Open* **2020**, *10*, e040132. https://doi.org/10.1136/bmjopen-2020-040132.
22. Shukla, V.V.; Eggleston, B.; Ambalavanan, N.; McClure, E.M.; Mwenechanya, M.; Chomba, E.; Bose, C.; Bauserman, M.; Tshefu, A.; Goudar, S.S.; et al. Predictive modeling for perinatal mortality in resource-limited settings. *JAMA Network Open* **2020**, *3*, e2026750. https://doi.org/10.1001/jamanetworkopen.2020.26750.
23. Gao, C.; Osmundson, S.; Yan, X.; Edwards, D.V.; Malin, B.A.; Chen, Y. Learning to Identify Severe Maternal Morbidity from Electronic Health Records. *Studies in Health Technology and Informatics* **2019**, *264*, 143–147.
24. Li, Q.; Alfonso, Y.N.; Wolfson, C.; Aziz, K.B.; Creanga, A.A. Leveraging Machine Learning to Predict and Assess Disparities in Severe Maternal Morbidity in Maryland. *Healthcare* **2025**, *13*, 284. https://doi.org/10.3390/healthcare13030284.
25. Nebraska Department of Health and Human Services, Division of Public Health, Office of Maternal and Child Health Epidemiology. Severe Maternal Morbidity Report 2017–2021. Technical report, Nebraska DHHS, 2024.
26. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **2011**, *12*, 2825–2830.
27. Chen, T.; Guestrin, C. XGBoost: A Scalable Tree Boosting System. In Proceedings of the Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, 2016, pp. 785–794.
28. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* **2002**, *16*, 321–357.
29. Breiman, L. Random Forests. *Machine Learning* **2001**, *45*, 5–32.
30. Geurts, P.; Ernst, D.; Wehenkel, L. Extremely Randomized Trees. *Machine Learning* **2006**, *63*, 3–42. https://doi.org/10.1007/s10994-006-6226-1.
31. Fawcett, T. An introduction to ROC analysis. *Pattern Recognition Letters* **2006**, *27*, 861–874.

32. Arlot, S.; Celisse, A. A survey of cross-validation procedures for model selection. *Statistics Surveys* **2010**, *4*, 40–79.
33. Riley, R.D.; Ensor, J.; Snell, K.I.E.; Harrell, F.E.; Martin, G.P.; Reitsma, J.B.; Moons, K.G.M.; Collins, G.; van Smeden, M. Calculating the sample size required for developing a clinical prediction model. *BMJ* **2020**, *368*, m441. https://doi.org/10.1136/bmj.m441.
34. Collins, G.S.; Moons, K.G.M.; Dhiman, P.; et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* **2024**, *385*, e078378. https://doi.org/10.1136/bmj-2023-078378.
35. Brier, G.W. Verification of forecasts expressed in terms of probability. *Monthly Weather Review* **1950**, *78*, 1–3.
36. Van Calster, B.; McLernon, D.J.; van Smeden, M.; Wynants, L.; Steyerberg, E.W. Calibration: the Achilles heel of predictive analytics. *BMC Medicine* **2019**, *17*, 230. https://doi.org/10.1186/s12916-019-1466-7.
37. Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K.Q. On Calibration of Modern Neural Networks. In Proceedings of the Proceedings of the 34th International Conference on Machine Learning, 2017, Vol. 70, *Proceedings of Machine Learning Research*, pp. 1321–1330.
38. Lundberg, S.M.; Lee, S.I. A unified approach to interpreting model predictions. In Proceedings of the Advances in Neural Information Processing Systems 30 (NeurIPS), 2017, pp. 4765–4774.
39. Lundberg, S.M.; Erion, G.; Chen, H.; DeGrave, A.; Prutkin, J.M.; Nair, B.; Katz, R.; Himmelfarb, J.; Bansal, N.; Lee, S.I. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence* **2020**, *2*, 56–67. https://doi.org/10.1038/s42256-019-0138-9.
40. Obermeyer, Z.; Powers, B.; Vogeli, C.; Mullainathan, S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* **2019**, *366*, 447–453. https://doi.org/10.1126/science.aax2342.
41. Rajkomar, A.; Hardt, M.; Howell, M.D.; Corrado, G.; Chin, M.H. Ensuring fairness in machine learning to advance health equity. *Annals of Internal Medicine* **2018**, *169*, 866–872. https://doi.org/10.7326/M18-1990.
42. Niculescu-Mizil, A.; Caruana, R. Predicting good probabilities with supervised learning. In Proceedings of the Proceedings of the 22nd International Conference on Machine Learning (ICML). ACM, 2005, pp. 625–632.
43. Mitchell, M.; Wu, S.; Zaldivar, A.; Barnes, P.; Vasserman, L.; Hutchinson, B.; Spitzer, E.; Raji, I.D.; Gebru, T. Model Cards for Model Reporting. In Proceedings of the Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). ACM, 2019, pp. 220–229. https://doi.org/10.1145/3287560.3287596.
44. Rieke, N.; Hancox, J.; Li, W.; Milletarì, F.; Roth, H.R.; et al. The future of digital health with federated learning. *npj Digital Medicine* **2020**, *3*, 119. https://doi.org/10.1038/s41746-020-00323-1.
45. Dayan, I.; Roth, H.R.; et al. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine* **2021**, *27*, 1735–1743. https://doi.org/10.1038/s41591-021-01506-3.
46. Vickers, A.J.; Elkin, E.B. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making* **2006**, *26*, 565–574. https://doi.org/10.1177/0272989X06295361.